MIL-B-37244A
14 May 1976
SUPERSEDING
MIL-B-37244
6 June 1975

MILITARY SPECIFICATION

BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Bupivacaine Hydrochloride and Epinephrine Injection.

1.2 Classification. The injection covered by this specification shall be the following strengths and sizes (see 6.2):

Strength 1 - 0.25% with Epinephrine 1:200,000
Strength 2 - 0.50% with Epinephrine 1:200,000
Strength 3 - 0.75% with Epinephrine 1:200,000

Size 1 - 50 ml
Size 2 - 30 ml, 5s
Size 3 - 50 ml, 5s

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

Federal

FED-STD-142	Parenteral Preparations.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS.

3.1 Material. Shall be Bupivacaine Hydrochloride and Epinephrine Injection and shall be in accordance with all requirements of FED-STD-142 and as specified herein:

S2 Classification. Shall be type I, class 1.

S5.2.2 The following is added to this paragraph:

For strength 1, shall be a sterile solution of Bupivacaine Hydrochloride, 0.25% with Epinephrine 1:200,000, within the designated assay limits for the injection.

For strength 2, shall be a sterile solution of Bupivacaine Hydrochloride, 0.50% with Epinephrine 1:200,000, within the designated assay limits for the injection.

For strength 3, shall be a sterile solution of Bupivacaine Hydrochloride, 0.75% with Epinephrine 1:200,000, within the designated assay limits for the injection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

Shall be suitable for caudal, epidural or peripheral nerve block.

3.1.1 Assay.

3.1.1.1 Bupivacaine Hydrochloride. The injection shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of Bupivacaine Hydrochloride, when determined as described in 4.3.1.1.

3.1.1.2 Epinephrine. The injection shall assay to contain not less than 90.0 percent and not more than 115.0 percent of the required amount of Epinephrine, when determined as described in 4.3.1.2.

3.1.2 Identification.

3.1.2.1 Bupivacaine Hydrochloride. The retention time of the Bupivacaine Hydrochloride shall match that of the reference standard similary treated (see 4.3.1.1).

3.1.2.2 Epinephrine. Shall be determined as specified in 4.3.1.3.

3.1.3 pH. The pH of the injection shall not be less than 3.4 and not more than 4.5 at 25°C. Determination shall be on the undiluted finished product.

3.2 Ingredients.

3.2.1 Bupivacaine Hydrochloride. The Bupivacaine Hydrochloride used in the manufacture of the finished injection shall comply with the following requirements:

3.2.1.1 Melting range. Shall melt between 247°C and 262°C with decomposition when determined using the U.S.P. Class Ia method without previous drying of the sample.

3.2.1.2 Heavy Metals. Shall not be more than 20 p.p.m when determined using the USP Method II.

3.2.1.3 Identification. The injection shall comply with the requirements of 4.3.2.1 and 4.3.2.2.

3.2.1.4 Assay. Shall assay not less than 98.5 percent and not more than 102.0 percent when determine as specified in 4.3.2.3.

3.2.2 Other ingredients. All other ingredients used in the manufacture of the injection shall comply with S5.1 of FED-STD-142.

3.3 Expiration dating period. The expiration dating period shall be not less than 24 months.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

3.3.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No compny supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-142.

4.3 Tests.

4.3.1 Finished product.

4.3.1.1 Bupivacaine Hydrochloride (Identification and Assay)

Gas Chromatographic Parameters: (will vary with instruments)

Gas Chromatograph: F & M Model 700 or equivalent
Column: 36 ft x 6 mm o.d. glass
Column Packing: 3% Hi-Eff-3BP (Neopentyl Glycol Succinate) on 100/120 mesh Gas Chrom Q
Column Temperature: 200°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

Injection Port Temperature: 250°C
Detector: Flame Ionization
Dectector Temperature: 250°C
Helium Cylinder Pressure: 30 psig
Air Cylinder Pressure: 15 psig
Hydrogen Cylinder Pressure: 22 psig
Attenuation: x 500

Standard Solution: Dissolve accurately weighed quantities of Bupivacaine Hydrochloride Reference Standard in purified water and dilute quantitatively with purified water to obtain a Standard Solution having approximately the same concentration as the sample to be analyzed.

Gas Chromatographic Assay: Inject exactly 5.0 µl of the Standard Solution onto the chromatographic column, and observe the peak for Bupivacaine Hydrochloride Reference Standard. Inject exactly 5.0 ul of the sample onto the chromatographic column, and observe the peak for Bupivacaine Hydrochloride. Calculate the quantity, in mg, of the component by the formula $C (A_U/A_S)$, in which C is the concentration , in mg per ml, of the component in the Standard Solution, and A_U and A_S are the chromatographic peak areas of the component in question in the sample and Standard Solution, respectively.

4.3.1.2 Epinephrine (Assay).

Reagents:

Purified Cation-Exchange Resin, sulfonic acid type, in the hydrogen form. The recommended grade is AG 50W-X1, 50-100 mesh (available from Bio Rad Laboratories, Richmond, Cal.) If equivalent resins are to be used, the acceptability must be verified by testing with appropriate epinephrine standard solutions.

Hydrochloric Acid 0.5N (Standardized Volumetric Solution)

Sodium Hydroxide 0.5N (Standardized Volumetric Solution)

Methyl Red Indicator Solution: Dissolve 100 mg of methyl red in 100 ml of alcohol, and filter, if necessary.

Sodium Acetate Solution 20%: Dissolve 66.4 gram of $NaC_2H_3O_2.3H_2O$ in 200 ml of purified water.

Potassium Ferricyanide-Sodium Acetate Solution: Dissolve 12.5 mg of $K_3Fe(CN)_5$ in 100 ml of 20% sodium acetate solution. Prepare just before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

Ascorbic Acid Solution 2%: Prepare just before use.

Alkaline Ascorbate Solution: Mix, just before use, 100 ml of 95% ethanol, 80 ml of 20% sodium hydroxide solution, and 8 ml of 2% ascorbic acid solution.

Preparation of Column: Place a small pledget of glass wool at the bottom of a glass column. Add purified water to the column, and slowly add the resin in the form of a slurry to a resin height of about 100 mm. Prior to use, pass 50 ml of 2.5N hydrochloric acid through the column, followed by 100 ml of purified water. Collect the final portion of effluent in a flask containing 10 ml of water and 1 drop of methyl red indicator solution. No red color should appear. Drain the water to the top of the resin column. Some batches of resin may not require the acid wash, only the wash with 100 ml of purified water. The analyst should determine whether the acid treatment is necessary by assaying appropriate epinephrine standard solutions.

Reference Solution: Prepare just before use. Dissolve 91.0 mg of epinephrine bitartrate USP reference stancard in 100.0 ml of purified water containing a drop of 6N hydrochloric acid. Dilute 10.0 ml to 100.0 ml with purified water containing a drop of 6N hydrochloric acid. Further dilute 10.0 ml to 100.0 ml with purified water containing a drop of 6N hydrochloric acid. The concentration of the final dilution is equivalent to 5.0 mcg of epinephrine base per ml.

Procedure: (NOTE: Thoroughly washed and rinsed glassware must be used in the fluorimetric procedure. Before use, rinse three times with distilled water.) Pipet 5.0 ml of sample into the column reservoir, partially opening the stopcock at the same time. Slowly drain the sample solution to the top of the resin column, then rinse the column reservoir with 5 to 10 ml of purified water. When this has slowly drained to the top of the resin, completely open the stopcock and wash the column with 75 to 100 ml of purified water, in 25-ml portions, draining each wash to the top of the resin. Discard the washes. Place a 50-ml volumetric flask under the column, and elute the epinephrine with two 25-ml portions of 0.5N hydrochloric acid at a flow rate of 0.5 to 1 ml per minute. (Total time required to collect the eluate must not be less than 60 minutes.) Collect exactly 50 ml of eluate, stopper the flask, and mix. Pipet 20.0 ml of the sample eluate into a 100-ml volumetric flask. Add 15.0 ml of 0.5N sodium hydroxide, swirling constantly during the addition. To each of four additional 100-ml volumetric flasks add 20.0 ml of 0.5N hydrochloric acid and 15.0 ml of 0.5N sodium hydroxide. Rinse

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

down the sides of the flasks with a small amount of purified water. Mix. Add, respectively, 0, 1.0, 2.0, and 3.0 ml of freshly prepared Epinephrine Standard Reference Solution (equivalent to 0, 5.0, 10.0,, and 15.0 mcg of Epinephrine. To each of the five sample and Standard flasks add 20.0 ml of freshly prepared potassium ferricyanide-sodium acetate solution. Swirl to mix. Exactly 2 minutes after the addition of the potassium ferricyanide-sodium acetate solution add 20.0 ml of freshly prepared alkaline ascorbate solution. Dilute to volume with water, and mix. Allow the solutions to stand for 15 minutes after the addition of the alkaline ascorbate solution. Determine the fluorescence intensity of the solutions in a spectrophotofluorometer (Aminco-Bowman SPF or equivalent), using an excitation wavelength of about 420 nm, a fluorescence wavelength of about 520 nm, and a slit arrangement No. 3. Plot corrected fluorescence versus concentration. Calculate the content of epinephrine in the sample from the Standard curve.

4.3.1.3 Epinephine (Identification)

Ferro-Citrate Solution (Doty Reagent): Dissolve 1.5 gram of ferrous sulfate ($FeSO_4.7H_2O$) in 200 ml of purified water containing 1 ml of 1N hydrochloric acid and 1 gram of sodium bisulfite, and dissolve 500 mg of sodium citrate in 10 ml of this solution.

Buffer Solution: Add 42 gram of sodium bicarbonate and 50 gram of potassium bicarbonate to about 180 ml of purified water. (not all of the solids will dissolve at this state.) To another 180 ml of purified water add 37.5 gram of aminoacetic acid and 17 ml of stronger ammonia water. Mix the two solutions, add purified water to make 500 ml, and mix until solution is complete.

Procedure: Transfer 10 ml of sample to a test tube, add 0.1 ml of Doty Reagent, and mix. Add 2 ml of Buffer Solution, mix, and allow to stand for 10 minutes. Filter through Whatman No. 1 paper. The filtrate is violet. Aged samples may produce a brown color in place of the violet color.

4.3.2 Ingredient (Bupivacaine Hydrochloride).

4.3.2.1 Infrared Spectrum. Dissolve about 230 mg in 15 ml of purified water in a separator. In a second separator dissolve about 230 mg of a reference standard in 15 ml of purified water. Treat each solution as follows: Add 1 ml of 10% ammonia solution and extract with three 30 ml portions of chloroform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

evaporate the chloroform at room temperature under a stream of nitrogen. Dry the residue in vacuum. Add 2.0 ml of chloroform, and mix to dissolve.

Determine the absorption spectra of the prepared solutions of both standard and sample, in 0.1 mm cells between 2 µm and 13 µm with a suitable infrared spectrophotometer, using chloroform in a matched cell as the blank. The spectrum of the solution prepared from the sample shall exhibit maxima only at the same wavelength as that of the reference standard.

A copy of the standard infrared spectrum shall be submitted by the contractor, and shall be kept on file in the Technical & Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, for use if samples are submitted.

4.3.2.2 Chloride. Dissolve about 50 mg of sample in 10 ml of purified water in a small separator. Make alkaline with 10% ammonia solution and extract with 10 ml of ether. Drain the aqueous layer into a test-tube and acidify with nitric acid. Add 0.1N silver nitrate. A white, curdy precipitate form which is insoluble in nitric acid but soluble in a slight excess of 10% amonia solution.

4.3.2.3 Assay. Accurately weigh about 600 mg of sample into a 250-ml conical flask. Dissolve in 20 ml of glacial acetic acid and 10 ml mercuric acetate solution (6% in glacial acetic acid). Add about 3 - 5 drops of crystal violet TS and titrate to an emerald green end point with 0.1N perchloric acid. Perform a blank determination and make any necessary correction.

1 ml 0.1N $HClO_4$ = 32.49 mg Bupivacaine Hydrochloride (anhydrous)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the ;unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3), and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 50 ml of strength 1, size 1 or strength 2, size 1;or one box containing five ampuls, each ampul containing 50 ml of strength 1, size 3;or 30 ml of strength 2, size 2 or strength 3, size 2, as specified constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available multiple dose vial for size 1 and a single dose ampul for size 2 and 3. The container shall be considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package.

5.2.2.2.1 Size 1. At the option of the contractor, a unit package may be furnished for each size 1 bottle. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.2.2 Size 2 and Size 3. Five ampuls of size 2 or size 3 of one strength only shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Five units of issue of one strength and size only shall be packaged in an intermediate package as specified in 5.2.8 of PPP-C-186.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Four intermediate packages (20 units of issue) of one strength and size only of strength 1, size 3; strength 2, size 2; strength 3, size 2; or twenty intermediate packages (100 units of issue) of strength 1, size 1 or strength 2, size 1 shall be packaged in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Four intermediate packages (20 units of issue) of one strength and size only of strength 1, size 3; strength 2, size 2; strength 3, size 2; or twenty intermediate packages (100 units of issue) of strength 1, size 1 or strength 2, size 1 shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

NOTE: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied for size 1 bottles, the National Stock Number shall appear on the immediate container label.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE."

5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

NATIONAL STOCK NO.	ITEM IDENTIFICATION	SPECIFICATION DESIGNATION
6505-00-149-3503	BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION, 0.25% with Epinephrine 1:200,000, 50 ml, 5s	Strength 1, Size 3
6505-00-149-3504	BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION, 0.25% with Epinephrine 1:200,000, 50 ml	Strength 1, Size 1
6505-00-149-3505	BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION, 0.75% with Epinephrine 1:200,000, 30 ml, 5s	Strength 3, Size 2
6505-00-149-3506	BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION, 0.50% with Epinephrine 1:200,000, 30 ml, 5s	Strength 2, Size 2
6505-00-149-3507	BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION, 0.50% with Epinephrine 1:200,000, 50 ml	Strength 2, Size 1

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1570

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-B-37244A BUPIVACAINE HYDROCHLORIDE AND EPINEPHRINE INJECTION

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:16Z
Check the source to verify that this is the current version before use.